

```
import pandas as pd
```

```
from google.colab import files
uploaded = files.upload()
```

Choose Files

No file chosen

Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.

Saving tested.csv to tested.csv

```
df=pd.read_csv('tested.csv')
df
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	892	0	3	Kelly, Mr. James	male	34.5	0	0	330911	7.8292	NaN	Q
1	893	1	3	Wilkes, Mrs. James (Ellen Needs)	female	47.0	1	0	363272	7.0000	NaN	S
2	894	0	2	Myles, Mr. Thomas Francis	male	62.0	0	0	240276	9.6875	NaN	Q
3	895	0	3	Wirz, Mr. Albert	male	27.0	0	0	315154	8.6625	NaN	S
4	896	1	3	Hirvonen, Mrs. Alexander (Helga E Lindqvist)	female	22.0	1	1	3101298	12.2875	NaN	S
...	...	...	...	...	...	...	...	...	...	...	...	...
413	1305	0	3	Spector, Mr. Woolf	male	NaN	0	0	A.5. 3236	8.0500	NaN	S
414	1306	1	1	Oliva y Ocana, Dona. Fermina	female	39.0	0	0	PC 17758	108.9000	C105	C
415	1307	0	3	Saether, Mr. Simon Sivertsen	male	38.5	0	0	SOTON/O.Q. 3101262	7.2500	NaN	S
416	1308	0	3	Ware, Mr. Frederick	male	NaN	0	0	359309	8.0500	NaN	S

Perform basic Exploratory Data Analysis (EDA) using functions like head(), tail(), describe(), shape.

```
df.head()
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	892	0	3	Kelly, Mr. James	male	34.5	0	0	330911	7.8292	NaN	Q
1	893	1	3	Wilkes, Mrs. James (Ellen Needs)	female	47.0	1	0	363272	7.0000	NaN	S
2	894	0	2	Myles, Mr. Thomas Francis	male	62.0	0	0	240276	9.6875	NaN	Q
3	895	0	3	Wirz, Mr. Albert	male	27.0	0	0	315154	8.6625	NaN	S
4	896	1	3	Hirvonen, Mrs. Alexander (Helga E Lindqvist)	female	22.0	1	1	3101298	12.2875	NaN	S

```
df.tail()
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
413	1305	0	3	Spector, Mr. Woolf	male	NaN	0	0	A.5. 3236	8.0500	NaN	S
414	1306	1	1	Oliva y Ocana, Dona. Fermina	female	39.0	0	0	PC 17758	108.9000	C105	C
415	1307	0	3	Saether, Mr. Simon Sivertsen	male	38.5	0	0	SOTON/O.Q. 3101262	7.2500	NaN	S
416	1308	0	3	Ware, Mr. Frederick	male	NaN	0	0	359309	8.0500	NaN	S

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 418 entries, 0 to 417
Data columns (total 12 columns):
#   Column             Non-Null Count  Dtype
---  ---
0   PassengerId         418 non-null    int64
1   Survived            418 non-null    int64
2   Pclass              418 non-null    int64
```

```

3   Name      418 non-null  object
4   Sex       418 non-null  object
5   Age       332 non-null  float64
6   SibSp     418 non-null  int64
7   Parch     418 non-null  int64
8   Ticket    418 non-null  object
9   Fare      417 non-null  float64
10  Cabin     91 non-null   object
11  Embarked  418 non-null  object
dtypes: float64(2), int64(5), object(5)
memory usage: 39.3+ KB

```

```
df.describe()
```

	PassengerId	Survived	Pclass	Age	SibSp	Parch	Fare
count	418.000000	418.000000	418.000000	332.000000	418.000000	418.000000	417.000000
mean	1100.500000	0.363636	2.265550	30.272590	0.447368	0.392344	35.627188
std	120.810458	0.481622	0.841838	14.181209	0.896760	0.981429	55.907576
min	892.000000	0.000000	1.000000	0.170000	0.000000	0.000000	0.000000
25%	996.250000	0.000000	1.000000	21.000000	0.000000	0.000000	7.895800
50%	1100.500000	0.000000	3.000000	27.000000	0.000000	0.000000	14.454200
75%	1204.750000	1.000000	3.000000	39.000000	1.000000	0.000000	31.500000
max	1309.000000	1.000000	3.000000	76.000000	8.000000	9.000000	512.329200

```
print('Number of rows and columns=',df.shape)
```

```
Number of rows and columns= (418, 12)
```

```
df.columns
```

```

Index(['PassengerId', 'Survived', 'Pclass', 'Name', 'Sex', 'Age', 'SibSp',
       'Parch', 'Ticket', 'Fare', 'Cabin', 'Embarked'],
      dtype='object')

```

Identify & handle missing value.

```
print(df.isnull()) #check for missing values
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	\
0	False	False	False	False	False	False	False	False	False	
1	False	False	False	False	False	False	False	False	False	
2	False	False	False	False	False	False	False	False	False	
3	False	False	False	False	False	False	False	False	False	
4	False	False	False	False	False	False	False	False	False	
..	...	...	...	...	...	...	...	...	...	
413	False	False	False	False	True	False	False	False	False	
414	False	False	False	False	False	False	False	False	False	
415	False	False	False	False	False	False	False	False	False	
416	False	False	False	False	True	False	False	False	False	
417	False	False	False	False	True	False	False	False	False	
	Fare	Cabin	Embarked							
0	False	True	False							
1	False	True	False							
2	False	True	False							
3	False	True	False							
4	False	True	False							
..	...	...	...							
413	False	True	False							
414	False	False	False							
415	False	True	False							
416	False	True	False							
417	False	True	False							

```
[418 rows x 12 columns]
```

```
print('Number of Missing values:',df.isnull().sum())
```

```

Number of Missing values: PassengerId      0
Survived              0
Pclass               0
Name                 0

```

```
Sex      0
Age      86
SibSp    0
Parch    0
Ticket   0
Fare     1
Cabin    327
Embarked 0
dtype: int64
```

```
#missing value found in column ["Age"],["Cabin"] and ["Embarked"]
#fill missing values in column age and embarked
df['Age'].fillna(df['Age'].mean(),inplace=True)
df['Embarked'].fillna(df['Embarked'].mode()[0],inplace=True)
```

/tmp/ipython-input-3337378054.py:3: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through chained assignment. The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are setting values is a copy. For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col] = df

```
df['Age'].fillna(df['Age'].mean(),inplace=True)
/tmp/ipython-input-3337378054.py:4: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through chained assignment. The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are setting values is a copy. For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col] = df

df['Embarked'].fillna(df['Embarked'].mode()[0],inplace=True)
```

```
print(df.isnull().sum())
```

```
PassengerId    0
Survived        0
Pclass         0
Name           0
Sex            0
Age            0
SibSp          0
Parch          0
Ticket         0
Fare           1
Cabin          327
Embarked        0
dtype: int64
```

```
df.drop(columns=['Cabin'],inplace=True)
print("Missing values handled")
```

Missing values handled

df

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Embarked
0	892	0	3	Kelly, Mr. James	male	34.50000	0	0	330911	7.8292	Q
1	893	1	3	Wilkes, Mrs. James (Ellen Needs)	female	47.00000	1	0	363272	7.0000	S
2	894	0	2	Myles, Mr. Thomas Francis	male	62.00000	0	0	240276	9.6875	Q
3	895	0	3	Wirz, Mr. Albert	male	27.00000	0	0	315154	8.6625	S
4	896	1	3	Hirvonen, Mrs. Alexander (Helga E Lindqvist)	female	22.00000	1	1	3101298	12.2875	S
...	...	...	...	...	...	...	...	...	...	...	...
413	1305	0	3	Spector, Mr. Woolf	male	30.27259	0	0	A.5. 3236	8.0500	S
414	1306	1	1	Oliva y Ocana, Dona. Fermina	female	39.00000	0	0	PC 17758	108.9000	C
415	1307	0	3	Saether, Mr. Simon Sivertsen	male	38.50000	0	0	SOTON/O.Q. 3101262	7.2500	S
416	1308	0	3	Ware, Mr. Frederick	male	30.27259	0	0	359309	8.0500	S

Check for duplicates and handle it.

```
#Detect outliers in 'fare'
Q1 = df['Fare'].quantile(0.25)
Q3 = df['Fare'].quantile(0.75)
IQR = Q3 - Q1
print('IQR=',IQR)
lower_bound = Q1 - 1.5 * IQR
upper_bound = Q3 + 1.5 * IQR
print('Lower bound=',lower_bound)
print('Upper bound=',upper_bound)
#Remove outliers
df = df[(df['Fare'] >= lower_bound) & (df['Fare'] <= upper_bound)]
print("Outliers handled")
```

```
IQR= 23.6042
Lower bound= -27.5105
Upper bound= 66.9063
Outliers handled
```

```
df.shape
```

```
(362, 11)
```

Count the feature value from suitable attribute.

```
#Count the feature values from column "Embarked"
Embarked_counts = df.groupby('Embarked').count()
print(Embarked_counts)
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	\
Embarked										
C	71	71	71	71	71	71	71	71	71	
Q	45	45	45	45	45	45	45	45	45	
S	246	246	246	246	246	246	246	246	246	

	Fare
Embarked	
C	71
Q	45
S	246

```
Sex_count = df.groupby('Sex').count()
print(Embarked_counts)
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	\
Embarked										
C	71	71	71	71	71	71	71	71	71	
Q	45	45	45	45	45	45	45	45	45	
S	246	246	246	246	246	246	246	246	246	

	Fare
Embarked	
C	71
Q	45
S	246

Apply data Encoding (Label Encoding, One hot encoding) technique on suitable attribute.

```
from sklearn.preprocessing import LabelEncoder
df_encoded = df.copy()
label_encoder = LabelEncoder()
df_encoded['Sex'] = label_encoder.fit_transform(df_encoded['Sex'])
df_encoded['Embarked'] = label_encoder.fit_transform(df_encoded['Embarked'])
df['class'] = label_encoder.fit_transform(df['Pclass'])
df_encoded
```

```
/tmp/ipython-input-126616674.py:6: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user\_guide/indexing.html#returning-a-view-versus-a-copy
df['class'] = label_encoder.fit_transform(df['Pclass'])
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Embarked	class
0	892	0	3	Kelly, Mr. James	1	34.50000	0	0	330911	7.8292	1	2
1	893	1	3	Wilkes, Mrs. James (Ellen Needs)	0	47.00000	1	0	363272	7.0000	2	2
2	894	0	2	Myles, Mr. Thomas Francis	1	62.00000	0	0	240276	9.6875	1	1
3	895	0	3	Wirz, Mr. Albert	1	27.00000	0	0	315154	8.6625	2	2
4	896	1	3	Hirvonen, Mrs. Alexander (Helga E Lindqvist)	0	22.00000	1	1	3101298	12.2875	2	2
...	...	...	...	...	...	...	...	...	...	...	...	...
412	1304	1	3	Henriksson, Miss. Jenny Lovisa	0	28.00000	0	0	347086	7.7750	2	2
413	1305	0	3	Spector, Mr. Woolf	1	30.27259	0	0	A.5. 3236	8.0500	2	2
415	1307	0	3	Saether, Mr. Simon Sivertsen	1	38.50000	0	0	SOTON/O.Q. 3101262	7.2500	2	2

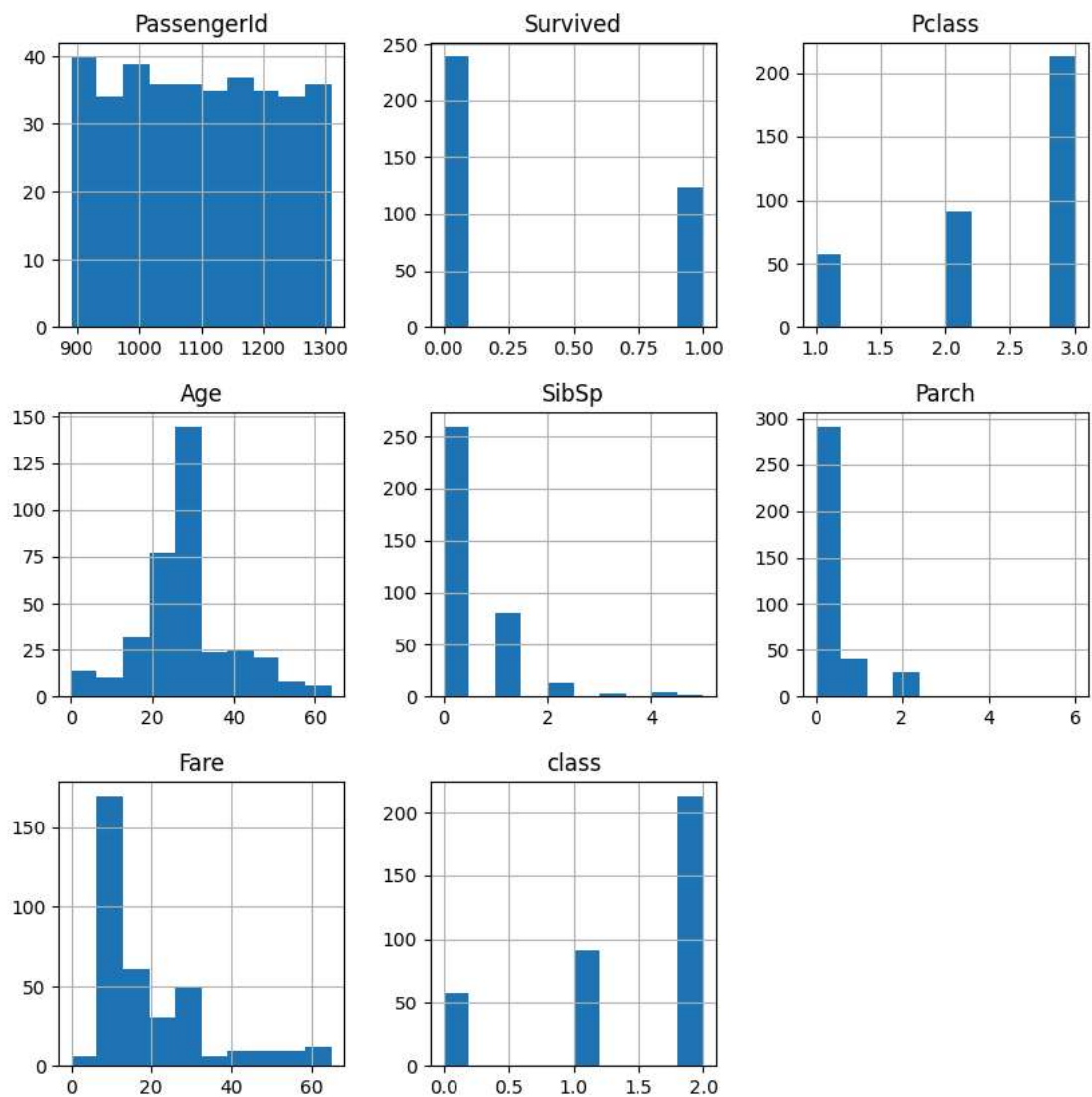
```
df_encoded = pd.get_dummies(df, columns=['Embarked'])
df_encoded
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	class	Embarked_C	Embarked_Q
0	892	0	3	Kelly, Mr. James	male	34.50000	0	0	330911	7.8292	2	False	True
1	893	1	3	Wilkes, Mrs. James (Ellen Needs)	female	47.00000	1	0	363272	7.0000	2	False	False
2	894	0	2	Myles, Mr. Thomas Francis	male	62.00000	0	0	240276	9.6875	1	False	True
3	895	0	3	Wirz, Mr. Albert	male	27.00000	0	0	315154	8.6625	2	False	False
4	896	1	3	Hirvonen, Mrs. Alexander (Helga E Lindqvist)	female	22.00000	1	1	3101298	12.2875	2	False	False
...	...	...	...	...	...	...	...	...	...	...	...	...	...
...	...	...	...	Henriksson, ...	...	...	...	...	...	...	...	...	...

Visualize distribution of individual features

```
import matplotlib.pyplot as plt
```

```
df.hist(figsize=(10,10))
plt.show()
```



```
df.boxplot(figsize=(7,7))
plt.title('Outlier Detection using Boxplot')
plt.show()
```

